Al-Balqa Applied University

Faculty of Engineering Technology, Basic Science Department, MID Exam

Statistics and prob. for eng. / Prob. and rand. processes

Instructor name:

Student Name:

Lecture Time:

Question Number	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	B	C	D	A	B	D	D	D	A	C

Question Number	Q11	Q12	Q13	Q14	Q15
Answer	✓	✓	✗	✓	✓

Question One: If $P(A|B) + P(B|A) = 0.8$ and

$$\frac{1}{P(B)} + \frac{1}{P(A)} = 4$$

Then $P(B \cap A) =$

A) 0.1 (B) 0.2 C) 0.3 D) 0.4 E) None

Question Two: For each positive integer n, we have $\binom{n}{0} =$

A) 0 B) $n!$ C) 1 D) undefined E) None

Question Three: A discrete random variable X with values $0, 1, -1$ with $P(X = 1) = P(X = -1) = c$. If $Var(X) = 4$; then $c =$

A) 0 B) 1 C) 3 D) 2 E) None

Question Four: If $P(A - \overline{B}) + P(A - B) = 0.1$, then $P(A) =$

(A) 0.1 B) 0.9 C) 0.2 D) 0.8 E) None

Question Five: Let f be a continuous function with $f(0) = 0$. If $\int_{-\infty}^{\infty} (2f(x-1) + f(x+1))\,\delta(x+1)\,dx = 4$: then $f(-2) =$

A) 1 B) 2 C) 3 D) 4 E) None

Question Six: Let X be a random variable with $E(X) = 1$ and $E(X^2) = 2$. We have $E(X^2 + X + Var(X)) =$

A) 1 B) 2 C) 3 D) 4 E) None

Question Seven: For continuous random variable X with cdf $F(x)$. If the cdf $F(x)$ is constant on the interval $[0,5]$ then find $P(0 < x < 2)$

A) 0 B) 1 C) 3 D) 2 E) None

Question Eight: Let X, Y be two independent random variables. If $Var(X) = 4$ and

$$(E(Y))^2 E(X^2) - (E(XY))^2 = 16$$

Then $|E(Y)| =$

A) 0.25 B) 1 C) 3 D) 2 E) None

Question Nine: For the joint PDF $f(x,y) = b^b$ when $0 < x < 1$, $0 < y < b^{-2}$ and zero otherwise, the positive constant $b > 1$ equals

A) 2 B) 4 C) 3 D) 5 E) None

Question Ten: Two independent events A, B with equal probability; if $P(A \cup B) = 0.5$; then $P(A) =$

A) $1 + \frac{1}{\sqrt{2}}$ B) $1 - \frac{1}{\sqrt{2}}$ C) $1 \pm \frac{1}{\sqrt{2}}$ D) $\sqrt{2}$

E) None

Question Eleven (✓, X) : A discrete random variable X with values $1, -1, 0$. We must have $E(X^2) = E(X^4)$

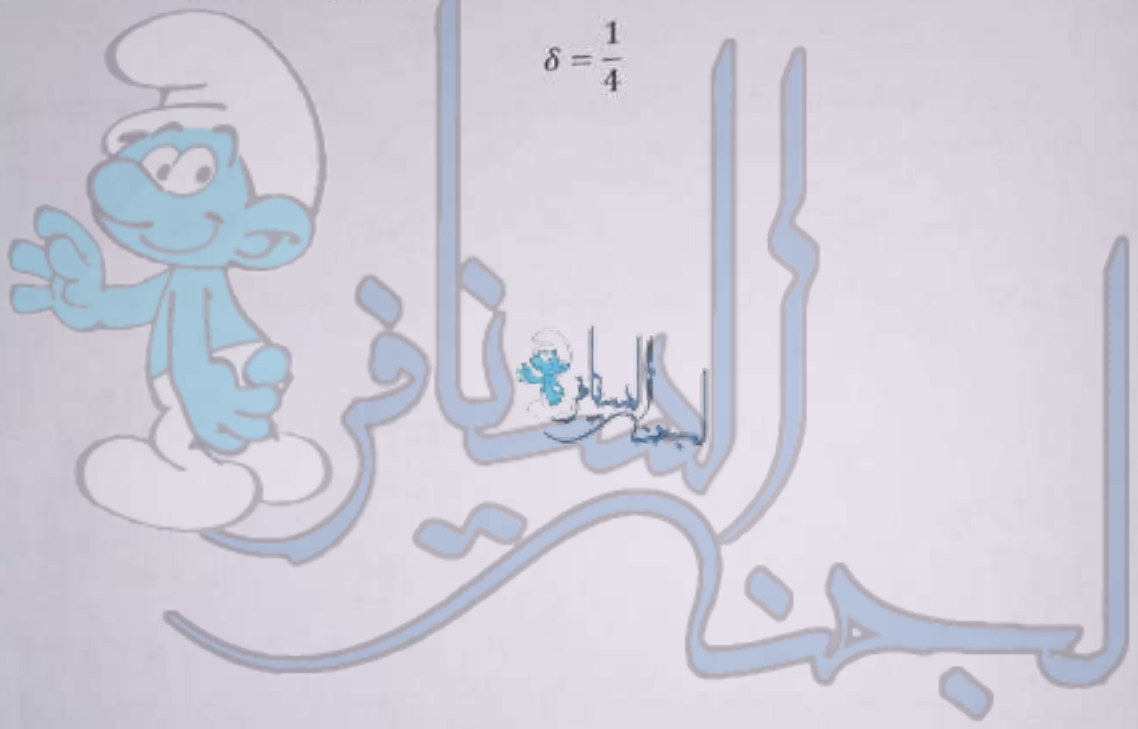
Question Twelve (✓, X) : For a random variable X with cdf $F(x) = 1 - 0.5e^{-2x}$, $x > -0.5\ln(2)$, the pdf has to be $f(x) = e^{-2x}$

Question Thirteen (✓, X) : When x is nonzero, we have $u(x) + u(-x) = 0$ ✗

Question Fourteen (✓, X): If $Cov(-X, -Y) = \sigma_X$ then $\sigma_Y > 1$ ✓

Question Fifteen (✓, X) : For the points of $R_{XY} = \{(a,a),(a,b),(b,a),(b,b)\}$ ✓
assume that $f(a,a) = f(b,b) = \delta$ and $f(a,b) = f(b,a) = 2\delta$. Then

$$\delta = \frac{1}{4}$$



Q1: $\frac{P(A \cap B)}{P(B)} + \frac{P(A \cap B)}{P(A)} = 0.8$

$P(A \cap B)\left(\frac{1}{P(B)} + \frac{1}{P(A)}\right) = 0.8$ (4)

$P(A \cap B) = \frac{0.8}{4} = 0.2$

Q2: $\binom{n}{0} = \frac{n!}{0!(n-0)!} = \frac{n!}{n!} = 1$

- remember: $0! = 1$

Q3: $Var(X) = EX^2 - (EX)^2$, $EX^2 = 4$ (0)

$EX = \sum_{x \in R_x} x p(x) = 0\,p(0) - 1\,p(-1) + 1\,p(1) = 0$ (0)

$4 = \sum_{x \in R_x} x^2 p(x) = 2c \longrightarrow c = 2$

$0 \leq p(x) \leq 1$ لكن السؤال فيه خطأ لأن

$p(1) = c = 2$?! وليس من المفترض أن

Q4: $P(A) - P(A \cap \bar{B})$ ← exception Law

$= P(A) - (P(A) - P(A \cap B)) + P(A) - P(A \cap B)$

$= P(A \cap B) + P(A) - P(A \cap B) = P(A) = 0.1$

Q5: $2\int_{-\infty}^{\infty} f(x-1)\,\delta(x-(-1))\,dx + \int_{-\infty}^{\infty} f(x+1)\,\delta(x-(-1))\,dx$

$= 2f(-1-1) + f(-1+1) = 4$

$2f(-2) = 4 \longrightarrow f(-2) = 2$

Q6: $Var(X) = EX^2 - (EX)^2 = 2 - (1)^2 = 1$

$E(X^2 + X + 1) = EX^2 + EX + 1 = 2 + 1 + 1 = 4$

Q7: $f(x)$ (p.d.f) $= \frac{dF(x)}{dx} = \frac{d}{dx} c = 0$

$f(x) = 0$ at $x \in [0,5]$ so $P(0 < x < 2) = 0$

Q8: $(EY)^2 EX^2 - (EX)^2 (EY)^2 = (EY)^2 (EX^2 - (EX)^2)$

$(EY)^2 Var(X) = 16 \quad (EY)^2 = 16/4$

$(EY)^2 = 4 \quad |EY| = 2$

Q9: $\iint_{R_{xy}} f(x,y)\,dx\,dy = 1$

$b^b \int_0^{b^{-2}} \left(\int_0^1 1\,dx \right) dy = 1 \rightarrow b^{b-2} = 1$

$b-2 \ln b = 0 \rightarrow b = 2$

Q10: $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$

but $P(A) = P(B)$

$\frac{1}{2} = 2P(A) - P^2(A)$

$P^2(A) - 2P(A) + 0.5 = 0$

$P(A) = 1 + \frac{1}{\sqrt{2}}$ or $1 - \frac{\sqrt{1}}{\sqrt{2}}$

$\rightarrow > 0 \leftarrow$

Q11: $EX^2 = (-1)^2 P(-1) + 0\,P(0) + P(1) = 2P(1)$

$EX^4 = (-1)^4 P(1) + 0\,P(0) + P(1) = 2P(1)$ ✓

Q12: $f(x) = \frac{d\,f(x)}{dx} = -0.5(-2)e^{-2x} = e^{-2x}$ ✓

Q13: $u(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$, $u(-x) = \begin{cases} 0, & x > 0 \\ 1, & x < 0 \end{cases}$

$u(x) + u(-x) = \begin{cases} 1+0, & x > 0 \\ 0+1, & x < 0 \end{cases}$ ✗

Q14: $Cov(-X, -Y) = (-1)(-1)\,Cov(X, Y) = 6X$ ~~+1~~

$$Corr(X, Y) = \frac{Cov(X, Y)}{6X \, 6Y} = \frac{1}{6Y}$$ ✓

Remember: $-1 \leq Corr(X, Y) \leq 1$

$6Y > 1$ ولكي هذا الرقم يجب أن يكون

Q15: $\sum_{x \in R_x} \sum_{y \in R_y} f(x, y) = 1$ ✗

$\big(f(a,a) + f(b,b)\big) + \big(f(a,b) + f(b,a)\big) = 1$

$2\delta \qquad\qquad 4\delta$

$6\delta = 1 \longrightarrow \delta = 1/6$